

LLM-Based Anonymization of Psychiatric Medical Records

Context-aware de-identification that removes extended PII while preserving clinical meaning.



Privacy-preserving · Clinically faithful

TEAM

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What we're building, and how we'll know it works

DATA & TASK

- 878 synthetic psychiatric intake records across 10 DSM-5 disorders
- Each record: therapist note + 10 personal Q&A + 10 clinical Q&A + span-tagged version
- Input: free text containing extended PII (ePII) — names, family, trauma, immigration, contact info
- Output: rewritten note with ePII removed, clinical content intact

NLP TASK TYPE

- Controlled text rewriting (not classification or extraction)
- Context-aware redaction via rephrasing, not entity masking
- 16 extended-PII categories — beyond the 18 standard HIPAA identifiers

EVALUATION STRATEGY



Tag-based validation

Exact-match check: were tagged PII spans removed? Were tagged clinical spans preserved?



QA-based validation (planned)

Re-ask the 20 personal/clinical questions against the anonymized text; score answer drift with BERTScore



Cross-model benchmark

Compare 3+ pretrained models under the same prompt, plus a prompting-strategy ablation

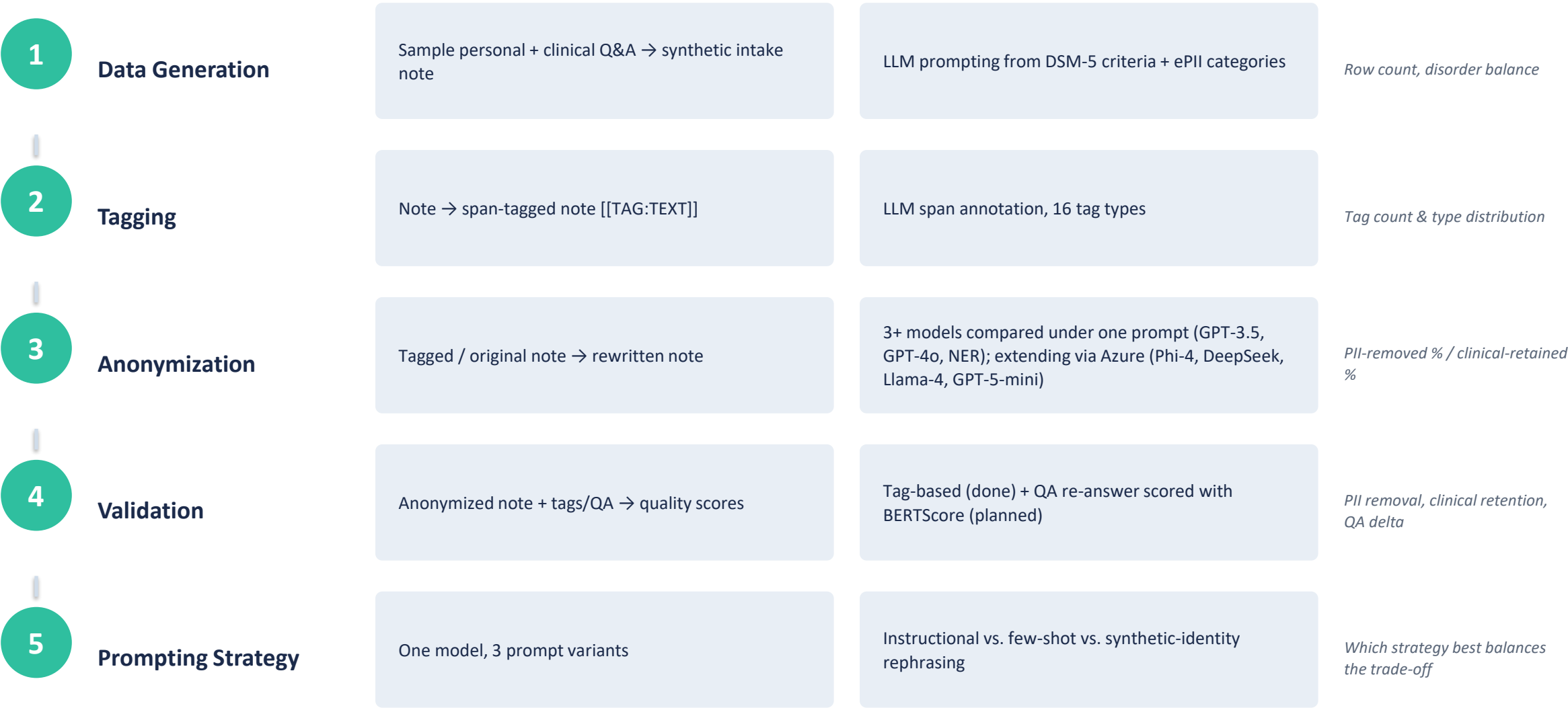
10

extended-PII tag types annotated per note, spanning identity, family, trauma, and immigration history

Where LLM-based de-identification stands today

Source	Approach	Data	Metric	Key result → how we differ
DeID-GPT (Liu et al., 2023)	GPT-4, zero-shot entity masking of PHI	General clinical notes	Precision / recall on PHI spans	High-precision masking; no fine-tuning required
Evaluating GPT models for de-identification (Sci. Reports, 2025)	GPT-3.5 / GPT-4 vs. clinician redaction	General clinical notes	Agreement with clinician redaction	Comparable to clinician redaction; no hallucination observed
LLMs-in-the-Loop: Expert Small Models (2024)	Specialized multilingual NER de-id models	Multilingual clinical text	PHI detection F1-score	F1 = 0.953–0.978; outperforms general LLMs on detection alone

Five stages, from synthetic data to validated anonymization



878 synthetic records, 3 baseline anonymization methods

878

patients, 10 disorders

172 ± 31

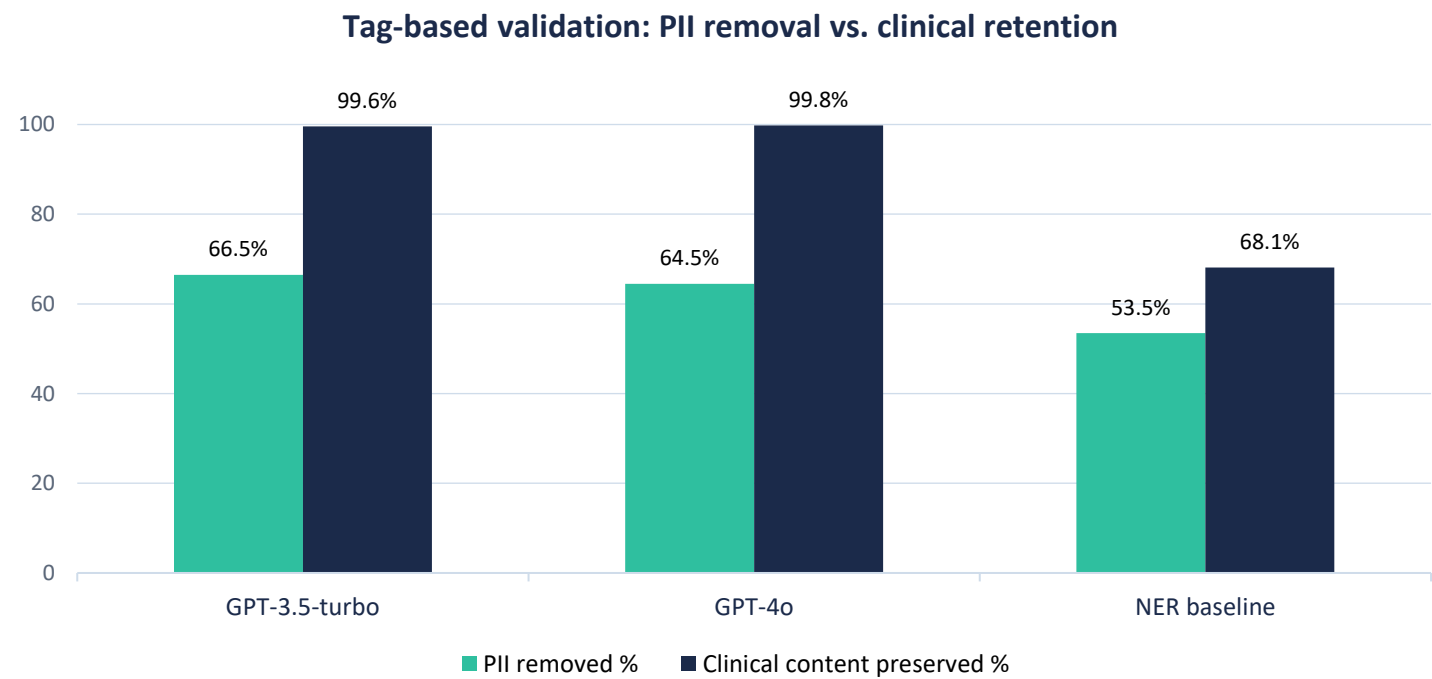
words per note

9,013

PII/clinical tags

79% / 21%

personal vs. clinical tags



READING THE CHART

- LLMs (GPT-3.5, GPT-4o) preserve clinical content almost perfectly (≥ 99.6%)
- But only ~65% of tagged PII is fully removed under strict exact-match scoring
- NER baseline masks entities blindly — worse on both axes, especially clinical retention (68.1%)
- 119/878 GPT-4o rows errored (API) — excluded from these percentages; noted as a limitation

What the baseline tells us, and where we go next

KEY INSIGHTS

- Rephrasing-based LLM anonymization preserves clinical meaning far better than entity masking (99.7% vs. 68.1%)
- PII removal (~65%) lags clinical retention — our exact-match scoring is strict, and the current prompt under-redacts some tagged spans
- CONTACT, NAME, and LOCATION dominate the personal-tag distribution — highest-value redaction targets
- GPT-4o's 13.6% error rate is an infrastructure issue (API), not a model-quality signal

NEXT STEPS

1

Fix GPT-4o error rows; scale the synthetic dataset further

2

Implement QA re-answering validation (re-ask CQ/PQ on anonymized text, score with BERTScore) as specified

3

Compare prompting strategies — few-shot and synthetic-identity rephrasing — on a fixed model

4

Broaden the model benchmark via Azure AI Foundry: Phi-4, DeepSeek, Llama-4, GPT-5-mini